

Instantaneous Traffic Generation from 5-Minute Measurements

Interrupted Poisson Process (IPP)

1 Introduction

The objective of this work is to generate synthetic instantaneous traffic (at the per-second scale) from aggregated 5-minute measurements using an Interrupted Poisson Process (IPP). The generated traffic must:

- preserve the empirical mean and variance of real data,
- reproduce realistic burstiness and idle periods,
- remain statistically consistent with cellular traffic dynamics.

2 Input Dataset and Notation

Traffic measurements are available every $T = 300$ seconds (5 minutes) for each sector. Let Ψ_i denote the measured traffic load (in Mbps) during the i -th interval.

The empirical statistics extracted from the data are:

$$\mu = \mathbb{E}[\Psi_i], \quad \sigma^2 = \text{Var}(\Psi_i) \quad (1)$$

These two quantities constitute the only constraints imposed on the synthetic traffic generation process.

3 IPP Traffic Model

3.1 ON/OFF Activity Process

Traffic activity is modeled as a two-state Markov chain:

$$X_t \in \{0, 1\},$$

where $X_t = 1$ corresponds to the ON state and $X_t = 0$ to the OFF state.

The transition probabilities per second are:

$$P(X_{t+1} = 0 \mid X_t = 1) = \tau, \quad (2)$$

$$P(X_{t+1} = 1 \mid X_t = 0) = \zeta. \quad (3)$$

The stationary probability of being in the ON state is:

$$P_{\text{ON}} = \frac{\zeta}{\tau + \zeta}. \quad (4)$$

3.2 Traffic Generation in ON State

When the system is ON, the number of arrivals per second follows:

$$U(t) \sim \text{Poisson}(\lambda). \quad (5)$$

Each arrival generates an independent traffic demand:

$$\psi_k \sim \text{Exp}(E[\psi]). \quad (6)$$

The instantaneous traffic load is:

$$\Psi(t) = \sum_{k=1}^{U(t)} \psi_k. \quad (7)$$

4 Moment Matching and Parameter Estimation

Using aggregation over a 5-minute interval, the feasibility constraint linking the empirical coefficient of variation and the IPP parameters is:

$$\frac{\sigma^2}{\mu^2} \geq \frac{2\zeta}{T\tau(\tau + \zeta)}. \quad (8)$$

The parameters (τ, ζ) are estimated numerically by minimizing:

$$\varepsilon = \left| \frac{\sigma^2}{\mu^2} - \frac{2\zeta}{T\tau(\tau + \zeta)} \right|. \quad (9)$$

Once (τ, ζ) are fixed, the remaining parameters are obtained in closed form:

$$\lambda = \frac{\tau + \zeta}{\tau T(r - r_{\min})}, \quad (10)$$

$$E[\psi] = \frac{(\tau + \zeta)\mu}{\tau \lambda T}, \quad (11)$$

where

$$r = \frac{\sigma^2}{\mu^2}$$

and

$$r_{\min} = \frac{2\zeta}{T\tau(\tau + \zeta)}.$$

As a result, the arrival rate λ varies across sectors and time intervals, reflecting differences in empirical traffic statistics.

4.1 Examples of Estimated Parameters

Table 1: Examples of estimated IPP parameters

Sector	τ	ζ	λ	$E[\psi]$
T36870A	0.05	1.00	0.150	4.00
T36870B	0.05	1.00	0.896	0.66
T70721C	0.05	1.00	1.84	0.41

For instance, sector T36870A yields $\lambda = 0.15$ and $E[\psi] = 4.0$, while denser sectors such as T70721C exhibit larger arrival rates ($\lambda = 1.84$) combined with smaller per-user traffic demands. This adaptive behavior allows the IPP model to reproduce both sparse and dense traffic regimes.

5 Instantaneous Traffic Generation Algorithm

For each 5-minute interval:

1. Initialize the ON/OFF state randomly.
2. For each second $t = 1, \dots, 300$:
 - Update the state using τ and ζ .
 - If OFF: $\Psi(t) = 0$.
 - If ON:
 - Draw $U(t) \sim \text{Poisson}(\lambda)$,
 - Generate $\Psi(t) = \sum_{k=1}^{U(t)} \psi_k$.

6 Statistical Validation: Sparsity Analysis

The global proportion of zero-valued instantaneous traffic samples is:

40.39%.

Table 2: Percentage of zero-valued instantaneous traffic samples by sector

Sector	Zero-valued samples (%)
T36870A	86.60
T36870B	90.06
T70721A	29.73
T70721B	13.31
T70721C	7.39
T78279B	4.70
T78279C	74.11
T78280A	66.52
T78280B	91.10
T78280C	43.52

The results reveal a strong heterogeneity in traffic activity across sectors. High sparsity levels indicate long OFF periods, while low sparsity reflects nearly continuous traffic activity. This diversity is naturally captured by the IPP ON/OFF dynamics.

7 Results and Figures

7.1 Normalized 5-Minute Load

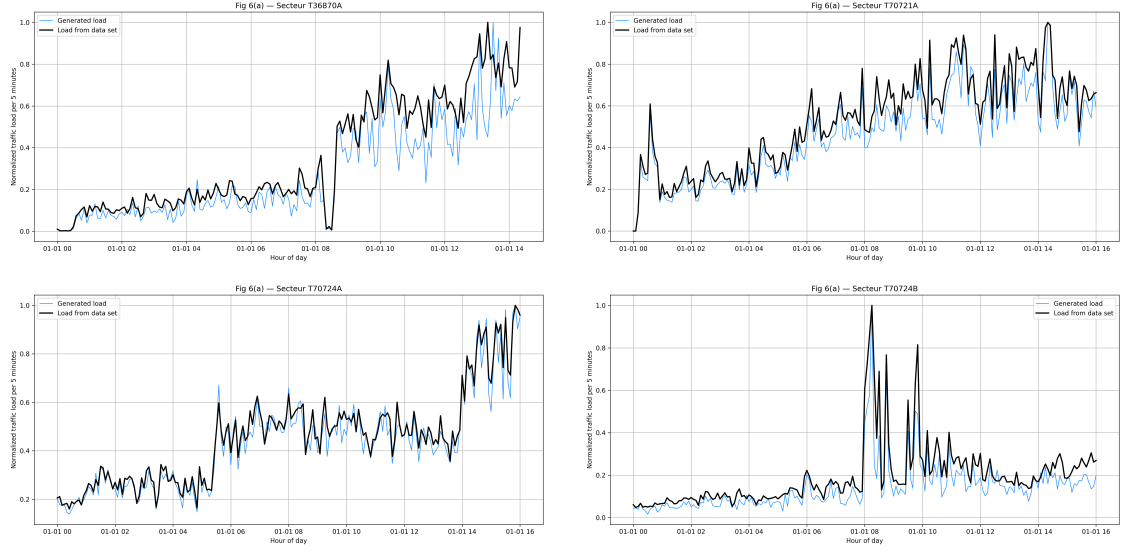


Figure 1: Normalized 5-minute traffic load for representative sectors

Figure 1 highlights distinct temporal traffic patterns across sectors. Despite normalization, sectors exhibit different peak intensities and daily profiles, motivating the use of sector-specific traffic modeling.

7.2 Instantaneous Traffic vs 5-Minute Load

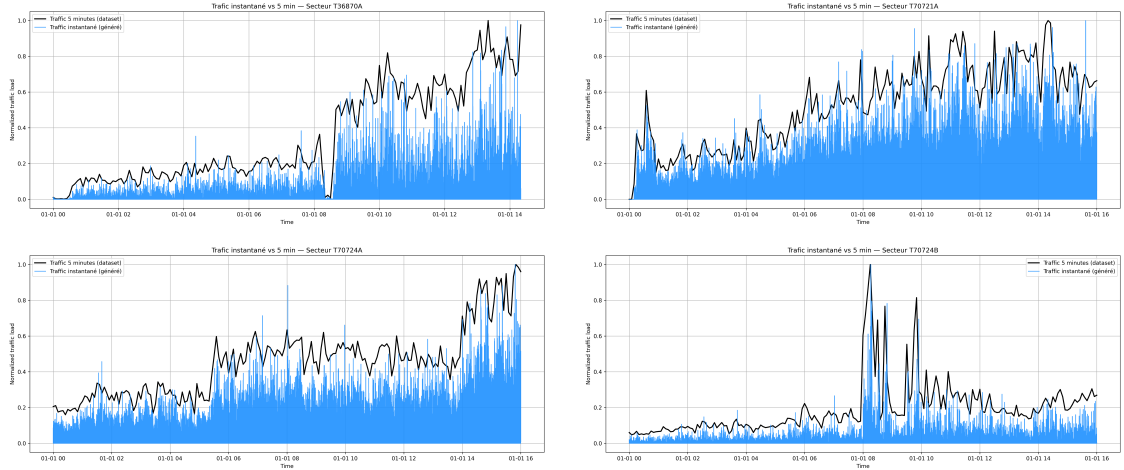


Figure 2: Instantaneous traffic versus 5-minute aggregated load

These figures show that the generated instantaneous traffic exhibits strong high-frequency fluctuations while preserving the envelope of the aggregated 5-minute measurements. When re-aggregated, the instantaneous traffic matches the original data, validating the consistency of the IPP-based generation process.